# MA323 : Lab 8 Report

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## Q1

We know that we can reduce the sample variance by using the control variate method. The required estimator is :

$$e^U + c(U - \mathbb{E}[U])$$

where

$$c = -\frac{Cov(e^U, U)}{Var(U)}$$

as this value of $c$ minimises the variance the most.
We can do the necessary computations and obtain this as our final estimator :

$$e^U - 6(3 - e)(U - 0.5)$$

### Results

The code was ran and following results were obtained :

| $M$ | $I_M$ | $Var$ | $\widehat{I}_M$ | $\widehat{Var}$ | %Imp | $\widetilde{I}_M$ | $\widetilde{Var}$ | %Imp |
|---|---|---|---|---|---|---|---|---|
| 10 | 1.56297 | 0.252897 | 1.71926 | 0.005398 | 97.866 | 1.72994 | 0.003525 | 98.606 |
| 100 | 1.73603 | 0.254168 | 1.72212 | 0.004182 | 98.355 | 1.72833 | 0.003827 | 98.494 |
| 1000 | 1.7215 | 0.228691 | 1.71687 | 0.00402 | 98.242 | 1.72011 | 0.003895 | 98.297 |
| 10000 | 1.71786 | 0.240794 | 1.71902 | 0.003936 | 98.365 | 1.71755 | 0.003881 | 98.388 |
| 100000 | 1.7179 | 0.242763 | 1.71833 | 0.003911 | 98.389 | 1.71823 | 0.003944 | 98.375 |

We can observe that both the methods can easily get upto 98% reduction in variance when compared to the crude technique. The control variate method also works slightly better than the antithetic variables method when less number of estimators ($M = 10, 100$) are used.

## Q2

For the conditioning technique, our estimator will be $\mathbb{E}[I_{(X>1)}|Y]$ and we know that $\mathbb{E}[I_{(X>1)}|Y = y] = 1 - \Phi(\frac{1-y}{2})$ since $X \sim N(y, 4)$
I don't know what control variable to take for the control variate technique. I tried taking $X$, $Y$ but the $c^*$ calculation doesn't look worth it. It's like cutting your arm in order to treat a fungal infection on your pinkie.
For the antithetic technique:

$$I_M = \frac{1}{M}\sum_{i=1}^{M/2} 1 - \frac{1}{2}[\Phi(\frac{1 + \ln(U_i)}{2}) + \Phi(\frac{1 + \ln(1 - U_i)}{2})]$$

### Results

$\widehat{I}_M$ denotes antithetic (using conditional variables) method and $\widetilde{I}_M$ denotes conditional method

| $M$ | $I_M$ | $Var$ | $\widehat{I}_M$ | $\widehat{Var}$ | %Imp | $\widetilde{I}_M$ | $\widetilde{Var}$ | %Imp |
|---|---|---|---|---|---|---|---|---|
| 10 | 0.2 | 0.177778 | 0.51552 | 0.041261 | 76.791 | 0.49834 | 0.02869 | 83.862 |
| 100 | 0.48 | 0.252121 | 0.48157 | 0.021341 | 91.535 | 0.49392 | 0.027611 | 89.049 |
| 1000 | 0.482 | 0.249926 | 0.48908 | 0.025813 | 89.672 | 0.48219 | 0.025807 | 89.674 |
| 10000 | 0.493 | 0.249976 | 0.48889 | 0.026009 | 89.595 | 0.49074 | 0.027101 | 89.159 |
| 100000 | 0.48946 | 0.249891 | 0.48992 | 0.026529 | 89.384 | 0.48914 | 0.026467 | 89.409 |

Both methods are at par I would say.

## Q3

It's pretty obvious that the discrete values of $N$ will be used as the stratas for this problem. The following results were obtained :

| $M$ | $I_M$ | $Var$ | $\widehat{I}_M$ | $\widehat{Var}$ | 99% CI |
|---|---|---|---|---|---|
| 100 | 0.38 | 0.23798 | 0.459999 | 0.001589 | (0.35714, 0.562858) |
| 10000 | 0.3663 | 0.232148 | 0.364099 | 0.000016 | (0.353633, 0.374566) |

The confidence interval given by the crude method was a super set of $[0, 1]$ so it didn't make sense to include it in the results. The variance reduction due to this method is drastic.